

HICKORY, NC, USA

WMO: 723010

Lat: 35.742N

Lon: 81.382W

Elev: 358

StdP: 97.10

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03810

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-4.8	-17.9	0.8	-1.9	-15.5	1.0	-1.3	8.9	7.4	8.1	7.0	2.0	320	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	33.3	22.6	32.2	22.4	31.0	22.1	24.5	29.8	24.0	29.0	23.4	28.2	2.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.6	26.0	22.5	18.0	25.7	22.1	17.5	25.4	75.7	30.0	73.5	29.1	71.5	28.2	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.6	5.5	-11.1	35.4	2.6	1.8	-12.9	36.7	-14.4	37.8	-15.8	38.8	-17.7	40.2		
			-12.1	25.4	2.3	0.7	-13.8	25.9	-15.2	26.4	-16.5	26.8	-18.1	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	4.3	6.3	10.3	15.4	19.7	23.8	25.2	24.8	21.5	15.6	10.1	6.1
	DBStd	8.40	5.37	4.70	4.97	4.29	3.48	2.52	2.10	2.21	3.11	4.13	4.41	4.59
	HDD10.0	575	188	121	60	7	0	0	0	0	0	6	53	139
	HDD18.3	1899	433	336	253	107	28	1	0	0	9	103	248	381
	CDD10.0	2513	13	18	68	169	300	416	472	458	346	180	56	18
	CDD18.3	796	0	0	2	18	70	167	214	200	106	18	1	1
	CDH23.3	6429	1	1	27	186	554	1386	1792	1600	740	134	8	1
	CDH26.7	2276	0	0	2	31	142	520	714	613	232	22	0	0
Wind	WSAvg	2.1	2.3	2.4	2.6	2.6	2.2	1.9	1.7	1.6	1.8	1.8	1.9	1.9
Precipitation	PrecAvg	1236	90	98	122	91	111	110	110	101	103	92	87	92
	PrecMax	1627	189	193	300	234	222	215	261	264	277	294	190	212
	PrecMin	837	11	11	34	3	26	16	2	12	1	0	22	5
	PrecStd	216	43	51	59	54	51	50	53	57	66	72	43	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.4	22.3	26.3	29.5	31.6	34.6	34.8	35.3	32.8	29.0	24.4	22.0
		MCWB	13.7	15.9	15.6	18.3	20.4	22.3	23.3	22.7	20.7	20.8	16.1	15.1
	2%	DB	17.6	19.1	23.5	27.4	29.8	32.6	33.1	32.9	31.2	26.6	21.7	18.6
		MCWB	13.4	12.7	14.6	17.1	20.2	22.1	23.3	22.7	21.0	18.6	14.5	14.5
	5%	DB	15.5	17.2	21.3	25.7	28.2	31.3	32.0	31.5	29.4	24.6	19.8	16.2
		MCWB	11.2	11.9	13.3	16.6	19.6	22.0	23.0	22.8	21.0	17.6	14.2	12.5
	10%	DB	12.9	14.7	18.8	23.6	26.6	29.8	30.7	30.1	27.7	22.6	17.8	13.6
		MCWB	9.2	9.3	12.1	15.6	19.0	21.4	22.7	22.3	20.3	17.0	13.2	9.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	17.3	17.6	20.4	22.8	24.5	25.5	25.2	23.9	23.0	19.3	17.9
		MCDB	18.5	20.3	23.2	25.7	27.8	30.3	31.4	30.5	28.3	25.7	20.6	20.0
	2%	WB	14.7	15.1	16.2	19.2	21.7	23.6	24.7	24.5	23.0	21.1	17.3	15.6
		MCDB	16.8	17.9	20.8	24.6	26.6	29.4	30.3	29.4	27.0	23.8	19.6	17.7
	5%	WB	12.3	12.8	15.1	18.0	21.0	23.0	24.1	23.9	22.3	19.6	15.8	13.1
		MCDB	14.3	16.0	19.3	22.9	25.9	28.7	29.2	28.5	26.2	22.9	18.5	15.2
	10%	WB	9.7	10.3	13.6	16.9	20.2	22.4	23.6	23.3	21.6	18.1	14.0	10.7
		MCDB	12.2	13.5	17.5	21.5	24.6	27.8	28.4	27.7	25.3	21.3	16.8	12.8
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.8	11.3	11.8	11.2	10.6	10.0	10.0	10.1	11.1	11.4	9.7
		MCDBR	13.0	14.2	14.0	14.1	13.2	12.7	11.7	11.9	12.5	13.1	13.8	12.6
		MCWBR	8.7	9.0	7.4	6.4	5.4	4.3	3.8	3.8	4.3	6.0	7.8	8.4
	5% WB	MCDBR	10.6	12.4	11.6	11.9	10.8	10.8	10.2	10.0	9.7	9.8	10.1	10.2
		MCWBR	8.4	9.3	7.4	6.2	5.0	4.2	3.9	3.8	4.1	5.6	7.6	8.0
Clear-Sky Solar Irradiance	taub	0.304	0.313	0.347	0.393	0.442	0.472	0.517	0.509	0.407	0.365	0.320	0.305	
	taud	2.512	2.480	2.425	2.297	2.183	2.136	2.038	2.039	2.310	2.417	2.506	2.539	
	Ebn at Noon	900	931	923	891	847	819	779	776	847	858	873	877	
	Edh at Noon	81	94	108	130	147	154	169	166	120	99	81	75	
All-Sky Solar Radiation	RadAvg	2.55	3.21	4.26	5.31	5.89	6.27	5.84	5.40	4.59	3.76	2.85	2.19	
	RadStd	0.16	0.36	0.29	0.41	0.47	0.45	0.41	0.26	0.42	0.46	0.24	0.21	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+131
(47)	Regional (56 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	+154

Nomenclature: See separate page